

# NovaPharma Iberia — Step 3.B Review: Production Pipeline

Digital transformation project · Excel → MySQL / Python / Power BI / Anaplan · Author: Nix · 21 September 2026

## What was done

The prototype notebook of Step 3.A became a production script that loads one monthly ERP extract into the warehouse, runs from a single terminal command, and is scheduled to run by itself. Four stages:

| Stage             | What                                                                                                                                                                                                      | Result                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 Schema          | Two operations tables added to the DDL: quarantine_sales (properly typed, with reason, source file and load time) and etl_log (one row per run)                                                           | 11 tables in novapharma_dw                               |
| 2 Script + config | etl_novapharma.py: six functions (config, extract, fetch dimensions, transform, load, log) chained by main. config.ini holds port, user and folders; the password lives in a Windows environment variable | Runs with: python etl_novapharma.py ERP_Sales_202609.csv |
| 3 Test            | Loaded September 2026 (1,574 read → 1,572 loaded, 2 quarantined). Ran the same file again: 0 loaded, 1,572 already present                                                                                | Idempotent: repeating does not duplicate                 |
| 4 Schedule        | Windows Task Scheduler task 'NovaPharma monthly ETL', day 3 at 07:00. Run manually once: etl_log run_id 10, OK, no human involved                                                                         | Monthly load automated                                   |

## The six functions, in plain words

| Function         | Does                                                                                                                                  |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| load_config      | Reads config.ini and opens the MySQL connection                                                                                       |
| extract_sales    | Reads the CSV; stops with an error if the 8 expected columns are not there (the macro's header check)                                 |
| fetch_dimensions | Gets the valid SKUs and customers from MySQL — the 'parents' every sale must have                                                     |
| transform_sales  | Applies the 8 cleaning rules of Step 3.A; rejected rows go to quarantine with a reason                                                |
| load_sales       | Checks which transaction IDs already exist, inserts only the new ones, adds the quarantine, marks the month as closed in the calendar |
| write_log / main | One row in etl_log per run (OK or FAIL); main calls everything in order inside a try/except and keeps a copy of the file in archive/  |

## Three ideas to remember

**Notebook designs, script runs.** The notebook keeps the exploration and the decisions; the script keeps only what is needed every month. Same rules, no describe(), no prints of discovery. It is run from the terminal because that is exactly what the scheduler will do.

**Parameters out of the code, secrets out of every file.** config.ini is the script's Parameters sheet: change a port or a folder without touching Python. The password is an environment variable: it is not in the code, not in the config, not in Git.

**Idempotent and logged.** A load that can be repeated safely, and that always leaves a trace — including when it fails — is what turns a macro into a pipeline. In Excel, someone had to press a button, change CurrentMonth and read Macro\_Log; now the task runs, closes the month and writes the log by itself.

## Backlog (known issues and improvements)

| # | Item                                                                                                                                             | Impact                                                     |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1 | Bug: quarantine rows are duplicated when the same file is re-run. A filter on (transaction_id, reason, source_file) was written but not verified | Low — quarantine is a register; loaded_at tells runs apart |
| 2 | Replace root/root with a dedicated etl_user limited to novapharma_dw                                                                             | Security                                                   |
| 3 | Detect the newest CSV in the inbox instead of fixing the file name in the scheduled task                                                         | Operations                                                 |
| 4 | Log distinguishes 'rejected' rows but not 'newly quarantined' ones                                                                               | Reporting                                                  |

## Vocabulary

Idempotent: running it twice gives the same result as running it once. Environment variable: a value stored in the user's Windows session, readable by any program, kept out of files. Task Scheduler (Programador de tareas): the Windows tool that runs a command on a calendar — unrelated to SQL window functions. def defines a function; nothing runs until something calls it, and main is what calls them all.